

2(a)

1 peat bog **A** has a higher number of species **and** a higher Shannon index (than peat bog **B**) ; **ora***examples of suitable wording for mp1:*

number of species	Shannon index
peat bog A has a higher number of species ora	peat bog A has a higher Shannon index ora
peat bog A has 12 species, peat bog B has <u>only</u> 8 species	<i>idea that</i> number of species / species richness, is more important for Shannon index
no (individuals of) species T, U, W, V in peat bog B	

2 peat bog **B** has a smaller range of species abundance **and** a higher Simpson's index (than peat bog **A**) ; **ora***examples of suitable wording for mp2:*

relative species abundance	Simpson's index
peat bog B has a smaller range of species abundance ora	peat bog B has a higher Simpson's index ora
peat bog B has a higher (total) number of individuals ora	<i>idea that</i> (range of) species abundance / species evenness / total number of individuals, is more important for Simpson's index
peat bog B has 135 (total) number of individuals, peat bog A has <u>only</u> 99	

*if mp1 and mp2 not awarded, then award 1 mark for:*peat bog **A** has a higher number of species **and** peat bog **B** has a smaller range of species abundance ; **ora**

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Question	Answer		Marks
2(a)	<i>examples of suitable wording:</i>		
	number of species	relative species abundance	
	peat bog A has a higher number of species ora peat bog A has 12 species, peat bog B has <u>only</u> 8 species no (individuals of) species T, U, W, V in peat bog B	peat bog B has a smaller range of species abundance ora peat bog B has a higher (total) number of individuals ora peat bog B has 135 (total) number of individuals, peat bog A has <u>only</u> 99	
2(b)(i)	<i>any three from:</i> 1 sample at different times of the year (not only over one month) ; 2 sample at, night / different times of day (not only between 13:00 and 16:00) ; 3 sample at sites other than along the path ; 4 use of (another) named technique to sample invertebrates ; 5 use an expert / (identification) app / guidebook (instead of an identification key) ;		3
2(b)(ii)	1 (for <i>Tipula limbata</i>) $n / N = 0.083$ and $(n / N)^2 = 0.007$; 2 $\Sigma = 0.270$; 3 $D = 0.730$;		3
2(b)(iii)	peat bog A has (slightly) higher (biodiversity than peat bog B) ; ora		1